

Binary Change Detection on EO-SAR Image Pairs

ABSTRACT

This project explores binary change detection using paired Electro-Optical (EO) and Synthetic Aperture Radar (SAR) imagery for disaster assessment. A UNet-based semantic segmentation framework with an EfficientNet-B0 encoder was implemented to identify damaged or destroyed regions between pre-event and post-event satellite imagery.

The dataset exhibited severe class imbalance, sparse foreground regions, and substantial modality differences between EO and SAR imagery. Initial experiments using BCE and Dice Loss resulted in model collapse toward background predictions. To address this, a weighted BCE strategy was introduced, significantly improving foreground sensitivity and recall.

The final pipeline demonstrates a complete end-to-end multimodal segmentation workflow including pre-processing, augmentation, training, validation, error analysis, and iterative improvement.

1. INTRODUCTION

Satellite-based disaster assessment is an important application of remote sensing and computer vision. Rapid identification of damaged infrastructure can significantly improve emergency response and resource allocation.

Binary change detection aims to identify regions that have changed between pre-event and post-event observations. In this work, multimodal EO and SAR imagery were used to improve robustness under varying environmental conditions.

EO imagery provides texture and color information, while SAR imagery captures structural and radar-reflectance properties. Combining both modalities presents challenges due to substantial differences in data distribution and noise characteristics.

2. DATASET ANALYSIS

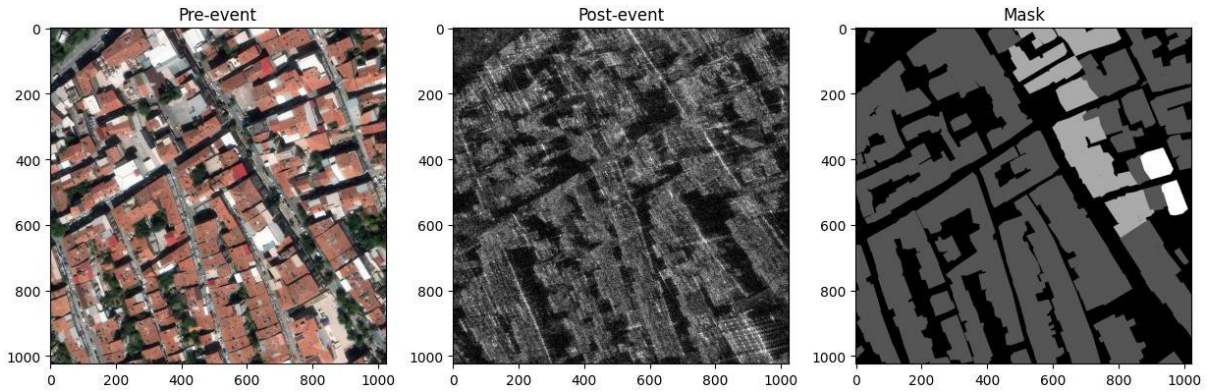


Figure 1. Example multimodal EO-SAR image pair and corresponding segmentation mask from the dataset.

The dataset contains paired pre-event and post-event images along with pixel-level masks.

Dataset structure:

```
train/
├── pre-event/
├── post-event/
└── target/
```

The original masks contained four semantic classes:

0 → Background
1 → Intact
2 → Damaged
3 → Destroyed

Binary remapping was applied:

0 → 0
1 → 0
2 → 1
3 → 1

Exploratory analysis revealed:

- severe class imbalance
- sparse foreground regions
- numerous tiles containing only background or intact structures
- SAR speckle noise
- strong EO↔SAR modality differences

These observations strongly influenced model and loss-function design.

3. METHODOLOGY

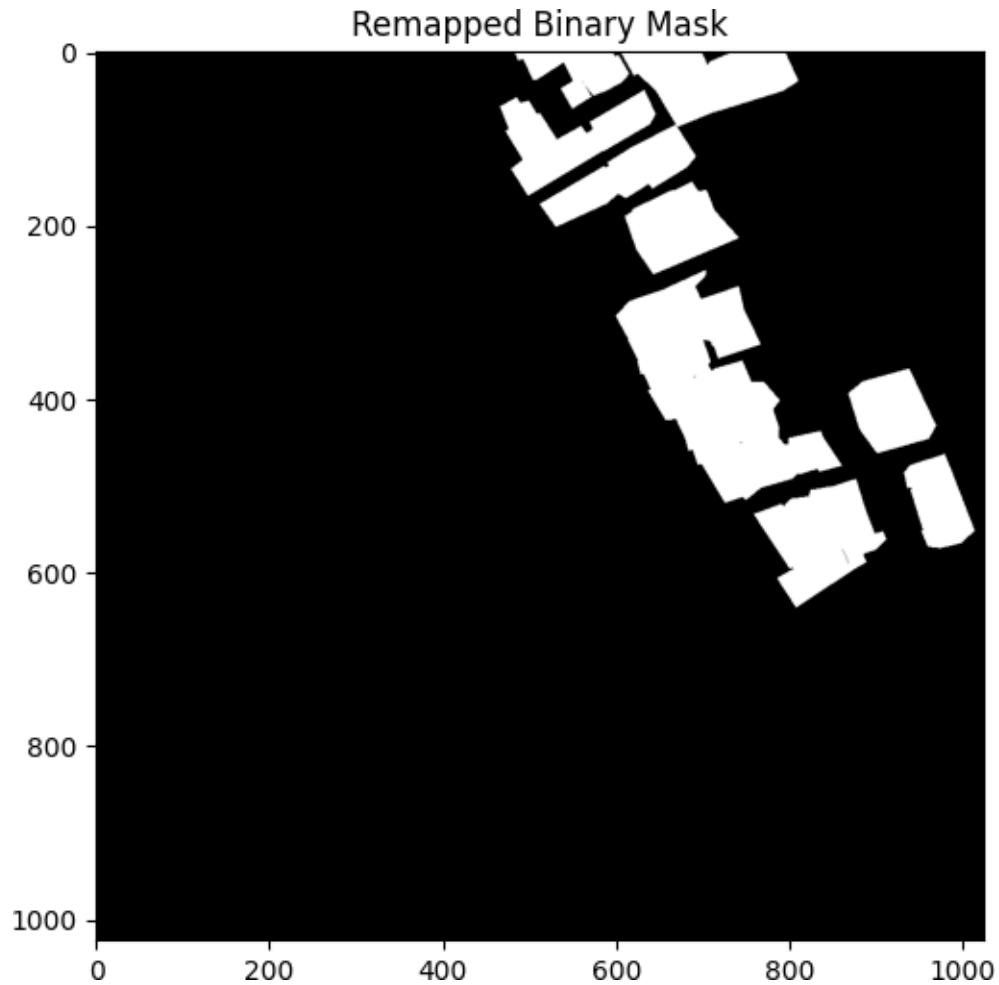


Figure 2. Binary remapping of original semantic labels into change/no-change classes.

Images were resized from 1024×1024 to 256×256 to balance computational efficiency and GPU memory constraints.

Data augmentations included:

- horizontal flips
- vertical flips
- random rotations
- brightness/contrast augmentation

The EO and SAR images were concatenated into a 6-channel tensor.

A UNet architecture with an EfficientNet-B0 encoder was used.

Reasons for selecting this architecture:

- strong segmentation baseline
- efficient training on limited GPU resources
- stable optimization
- Lightweight encoder suitable for Colab GPU constraints.

Two experiments were conducted.

Experiment 1:

- BCEWithLogitsLoss
- Dice Loss

Problem observed:

- model collapsed toward predicting only background regions

Experiment 2:

- Weighted BCEWithLogitsLoss
- Dice Loss

A positive-class weight was introduced to increase sensitivity toward sparse foreground pixels.

Optimizer:

- Adam

Learning rate:

- $1e-4$

Batch size:

- 4

Epochs:

- 5

4. RESULTS

Two experiments were conducted during training.

Experiment 1

The initial model used BCEWithLogitsLoss combined with Dice Loss. Due to severe class imbalance, the model collapsed toward predicting only background regions.

Results:

- Best Validation IoU: 0.4405
- Precision: 0.0000
- Recall: 0.0000
- F1 Score: 0.0000

Observation:

- the model produced mostly empty masks
- foreground regions were not detected successfully

Experiment 2

To address foreground sparsity and imbalance, weighted BCEWithLogitsLoss was introduced using:

`pos_weight = 8.0`

This significantly improved sensitivity toward damaged regions.

Results:

- Validation IoU: 0.1427
- Precision: 0.0178
- Recall: 0.4141
- F1 Score: 0.0341

Observation:

- recall improved substantially
- the model detected more foreground regions
- however, false positives increased significantly.

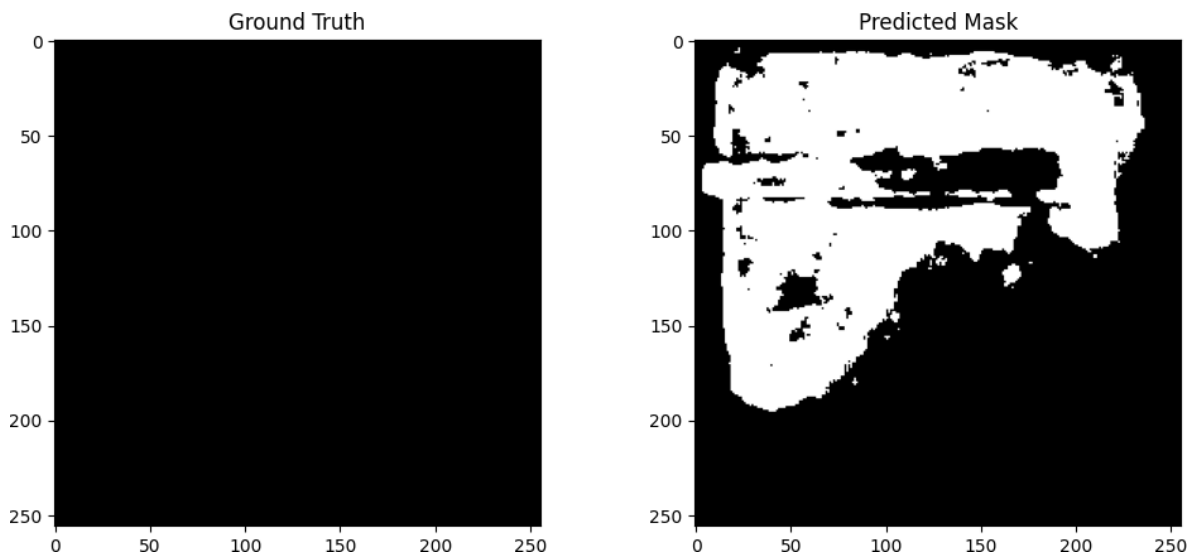


Figure 3: Qualitative comparison between the ground truth mask and the predicted mask after weighted BCE training.

5. ERROR ANALYSIS

The experiments highlighted multiple challenges.

1. Severe Class Imbalance

Many samples contained few or no positive pixels, causing unstable optimization.

2. Background Collapse

The initial model learned to predict only background regions because this minimized loss effectively.

3. False Positive Expansion

Weighted BCE improved recall but increased noisy predictions.

4. EO-SAR Modality Gap

EO and SAR imagery exhibit fundamentally different appearance distributions, making feature fusion difficult.

6. FUTURE WORK

Potential improvements include:

- Focal Loss
- Tversky Loss
- Attention-based UNet architectures
- Transformer-based fusion networks
- Better class balancing strategies
- Multi-scale training
- Threshold optimization
- Self-supervised pretraining
- Advanced SAR denoising

7. CONCLUSION

This work implemented a complete multimodal semantic segmentation pipeline for binary change detection using EO and SAR imagery.

The experiments demonstrated the significant impact of class imbalance and sparse foreground regions in disaster assessment tasks. Weighted loss functions improved foreground sensitivity but introduced increased false positives.

Despite modest final metrics, the project successfully demonstrated:

- multimodal pre-processing
- segmentation model training
- experimental debugging
- iterative improvement
- understanding of remote sensing challenges

8. TIME AND RESOURCE LOG

Approximate time spent:

- Dataset exploration: 2 hours
- Literature review: 2 hours
- Implementation: 5 hours
- Training and evaluation: 3 hours
- Debugging and experimentation: 3 hours
- Report writing: 2 hours
- Training Environment:
- Google Colab
- NVIDIA Tesla T4 GPU
- Approximate training time per epoch: 10–15 minutes
- Batch size: 4
- Image size: 256×256